

Khulna University of Engineering & Technology
B.Sc. Engineering 1st Year 1st Term (Regular) Examination, 2018
Department of Materials Science and Engineering

Ph 1127
Optics and Waves

Time: 3 hours

Full Marks: 210

- N.B. (i) Answer ANY THREE questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer ANY THREE questions from this section in answer script A)

- Q 1. (a) Show that the superposition principle is valid only in the case of homogeneous linear vibrations. (10)
- (b) Let the displacement of a particle executing simple harmonic motion at any instant is given by, $y = a \sin(\omega t + \delta)$. Hence show that the total energy of a particle is proportional to the square of the amplitude of the motion. Also show that the principle of conservation of energy is obeyed by a harmonic oscillator. (15)
- (c) A massless spring, suspended from a rigid support, carries a mass of 600 gm at its lower end and the system oscillates with a frequency of 6 Hz. If the amplitude is reduced to half its undamped value in 22 sec, calculate (i) the force constant of the spring, (ii) the relaxation time of the system and (iii) its quality factor. (10)
- Q 2. (a) Explain clearly the terms: free vibration, forced vibration and resonance. Give example of each. (10)
- (b) Explain analytically the formation of stationary waves. How can you indicate the positions of nodes and antinodes from the above equation? (15)
- (c) Plane harmonic waves of frequency 500 Hz are produced in air with displacement amplitude of 10^{-3} cm. Deduce (i) the pressure amplitude, (ii) energy density and (iii) energy flux in the wave. (10)
- Q 3. (a) What is Doppler's effect? Find the expression for the apparent frequency of the note when the source and listener are (i) moving towards each other and (ii) moving away from each other. (15)
- (b) Define the intensity and energy density at a point in a plane wave of sound. Obtain an expression for the intensity of a plane wave in terms of acoustic pressure. (10)
- (c) The apparent frequency of the whistle of an engine changes in the ratio 6:5 as the engine passes a stationary observer. If the velocity of sound is 340 m/s, calculate the velocity of the engine. (10)
- Q 4. (a) What is meant by ultrasonic waves? Mention the properties of ultrasonic waves. (10)
- (b) Explain what is meant by bel and phon. Define the terms (i) Threshold of feeling, (ii) Threshold of hearing, (iii) Band power level, (iv) Peak level and (v) Noise reduction (15)
- (c) A room dimensions $12 \times 10 \times 10$ cubic meters. Calculate (i) the mean free path of the sound wave in the room (ii) the number of reflections made per second by the sound wave with the walls of the room. Velocity of sound in air 340 m/s. (10)

Section B

(Answer ANY THREE questions from this section in answer script B)

- Q 5. (a) Describe the construction of a compound microscope and explain its action with a simple ray diagram. (15)
- (b) What are chromatic and spherical aberration for a lens? Describe without mathematical details, how these defects can be reduced? (10)
- (c) Two thin convex lenses having focal length of 5 cm and 2 cm are co-axial and separated by a distance of 3 cm. Find the equivalent focal length and the position of the principal points. (10)
- Q 6. (a) What is meant by interference of light? Discuss the conditions for interference. (05)
- (b) Describe Young's double slit experiment and hence show that the distance between two consecutive bright fringes formed in this experiment is given by, $\beta = \lambda D/d$. (10)
- (c) According to the Young's double slit experiment, the resultant intensity distribution due to the interference at any point is, $I = 4a^2 \cos^2 \left(\frac{\delta}{2} \right)$. Hence, (10)
- (i) Find the condition for constructive interference
- (ii) Find the condition for destructive interference
- (iii) Also show that, the formation of interference fringes is in accordance with the law of conservation of energy.
- (d) In a Newton's rings experiment the diameter of the 15th ring was found to be 0.590 cm and the 5th ring was 0.336 cm. If the radius of the plano-convex lens is 100 cm, calculate the wavelength of the light used. (10)
- Q 7. (a) What is diffraction? On what factors "double-slit diffraction" differ from "single-slit diffraction"? (10)
- (b) Show that, the intensity distribution for single slit Fraunhofer diffraction can be expressed as, $I = I_0 \left(\frac{\sin^2 \alpha}{\alpha^2} \right)$, where I_0 is the central maximum (maximum intensity, $\theta=0$). (15)
- (c) A grating has 1000 lines ruled on it. In the region of wavelength $\lambda = 6000 \text{ \AA}$, find (i) the difference between two wavelengths that just appear separated in the first order and (ii) the resolving power in the second order spectrum. (10)
- Q 8. (a) Explain the following terms: (i) Unpolarized light, (ii) Polarized light, (iii) Polarization of light, (iv) Plane of polarization and (v) Plane of vibration (10)
- (b) State Brewster's law. Show from this law that when the light is incident on a transparent substance at the polarizing angle, the reflected and refracted rays are at right angles. (15)
- (c) A solution of an optically active solute produces a rotation of 20° of the plane of polarization in a path length of 10 cm when the concentration is 20 gm/liter. What is the concentration in a solution which produces a rotation of 30° in a path length of 5 cm? (10)

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MSE 1101

Introduction to Materials Science and Engineering

Time: 3 hours

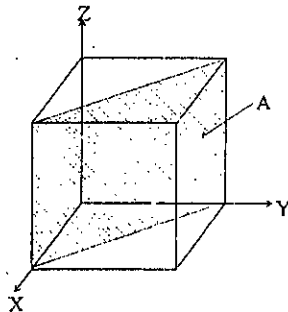
Full Marks: 210

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Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

- Q 1. (a) Draw and illustrate schematic diagram of bonding energy and interatomic separation and write equation for net potential energy. (13)
- (b) Explain, why hydrogen fluoride (HF) has a higher boiling temperature than hydrogen chloride (HCl) even though HF has a lower molecular weight? (12)
- (c) Calculate the force of attraction between Ca^{2+} and O^{2-} ions, the center of which are separated by distance of 1.25 nm. (10)
- Q 2. (a) Write down the types of cubic crystal system. Draw a Face center crystal system and calculate the total number of atoms in this system. (13)
- (b) Determine c/a ratio of a hexagonal closed packed structure. (12)
- (c) Describe the steps in determining the miller indices of a crystallographic plane. What is the miller indices of the plane A drawn below: (10)



- Q 3. (a) Describe the electrical conductivity of different materials based on energy band diagram. (12)
- (b) Explain, why the electrical conductivity of a semiconductor increases with increase in temperature while the conductivity decreases in metals? (10)
- (c) Compute the electrical conductivity of a 7 mm diameter cylindrical specimen of 57 mm long in which a current of 0.25 A passes in an axial direction. A voltage of 24 V is measured across two probes that are separated by 45 mm. Compute the resistance over the entire 57 mm of the specimen. (13)
- Q 4. (a) What is ferromagnetism? Draw the magnetic moment diagram of three types of materials with and without application of external magnetic field. (13)
- (b) A brass rod is stress free at room temperature (20 °C). It is heated up, but prevented from lengthening. At what temperature does the stress reach -172 MPa? (Coefficient of thermal expansion $20 \times 10^{-6}/^\circ\text{C}$) (10)
- (c) What is NDT? Describe the working principle of ultrasonic testing method and write down its advantages and limitations. (12)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

Q 5. (a) A better combination of properties can be achieved by the judicious combination of two or more distinct materials while developing a composite materials—explain. (10)

(b) In an aligned and continuous carbon fiber reinforced nylon 6,6 composite, the fibers are to carry 97% of a load applied in the longitudinal direction. (12)

(i) Using the data provided, determine the volume fraction of fibers required

(ii) What is the tensile strength of this composite? Assume the matrix stress at fiber failure is 50 MPa.

Given,

	<u>Modulus of Elasticity, GPa</u>	<u>Tensile Strength, MPa</u>
Carbon fiber :	260	4000
Nylon 6,6 :	2.8	76

$$V_m + V_f = 1; \sigma_c = \sigma_m \sigma_m + \sigma_f \sigma_f; F_c = F_m + F_f; \frac{F_f}{F_m} = \frac{E_f V_f}{E_m V_m}; \sigma_{cl} = \sigma'_m V_m + \sigma'_f V_f$$

symbols are indicating their usual meaning.

(c) Write few common properties of ceramics. Briefly explain at what condition it is advantageous to use ceramics over metallic materials. (13)

Q 6. (a) What are the common reasons of materials failure? What is your role as a Materials Engineer to impede this failure? (10)

(b) Explain the ductile failure of materials. Describe the stages in cup-and-cone fracture during ductile failure. (12)

(c) What is creep failure? Explain the mechanism of creep failure with the help of a creep curve. (13)

Q 7. (a) Explain why engineering stress and strain are used in conventional process instead of true stress and strain of a material. (10)

(b) Write the names of crystal defect that are commonly found in crystalline solids. Explain any one of them. (13)

(c) Write short notes on optical properties of metals. Why do Aluminum (Al) and Silver (Ag) exhibits bright-silvery color whereas Copper (Cu) exhibits red-orange color, explain. (12)

Q 8. (a) A material can be made stronger by applying plastic deformation—explain with the help of strain hardening mechanism. (10)

(b) How does solid-solution strengthening process increase the strength of materials? (15)

(c) Briefly explain the mechanism of plastic deformation by dislocation movement through a single crystal. (10)